Youssof Ahmed Aly

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Summary

Computer Science graduate with a dual degree from the University of Northampton (UK) and the Arab Academy for Science, Technology and Maritime Transport (Egypt). Cybersecurity & AI intern at MAHE–ISAC Centre of Excellence, Manipal, India; PPO reinforcement-learning graduation project; top 20% at the DeepX AI Competition (30 of 150 teams). Immediately available for cybersecurity, data science/ML and software engineering roles.

Education

B.Sc. Computer Science — University of Northampton (UK) Oct 2022 – Jul 2026

Dual award with the Arab Academy for Science, Technology & Maritime Transport, Egypt

Coursework: Cybersecurity, Computer System Security, Deep Learning, NLP, Digital Image Processing, Artificial Intelligence, Computer Networks, Operating Systems, Data Structures & Algorithms, Database Systems

High School Diploma (IGCSE) — Ethos International School 2017 – 2022

Experience

Cybersecurity & AI Intern — MAHE–ISAC Centre of Excellence, Manipal, India (via AIESEC) Jul – Sep 2025

- Built a fingerprint identification system (Python, image processing) and a machine-learning signature verification model.
- Developed a secure file upload service with MIME validation, ClamAV integration, HMAC-signed downloads and per-IP rate limiting.
- Mentored by Dr. Srikanth Prabhu (SoCE); received an official completion certificate from MIT Manipal.

Teaching Assistant — Ethos International School Sep 2022 – Jan 2023

- Supported classroom management and gave academic guidance to students needing extra support.

Projects

Selected academic, internship and personal builds across cybersecurity, machine learning and systems programming.

AI-Driven HPC Cluster Scheduler — Graduation project, team of 5 · Python, FastAPI, Celery, Redis, React, TypeScript, Stable-Baselines3
PPO reinforcement-learning agent that learns job-placement policies to cut wait time and raise node utilisation, with a live React dashboard.

Fingerprint Identification System GitHub ↗ · Python, OpenCV, scikit-learn
Classifies fingerprint patterns (arch, loop, whorl) using image processing and ML.

Secure File Upload Service · Python, FastAPI, Docker, ClamAV
Vulnerable vs. hardened upload demo: MIME sniffing, ClamAV scanning, HMAC-signed downloads, rate limiting.

Cryptographic Protocol Analyzer · Python, Scapy, Streamlit, Pandas, Plotly
Rule-based PCAP analyzer that flags insecure protocols, deprecated TLS versions and plaintext credentials, rated Critical to Low.

Virtual File System GitHub ↗ · Python, SSH
File system with a CLI shell, GUI and SSH remote access.

SysMonitor GitHub ↗ · Python, Bash, Docker, Chart.js
Real-time CPU/GPU/temperature monitor with Markdown reports and a live web dashboard.

Evil Dungeons GitHub ↗ · C++, GLSL, CMake
Dungeon-crawler with custom GLSL shaders and low-level graphics programming.

Technical Skills

Languages: Python, C, C++, Bash, JavaScript, TypeScript, SQL
Cybersecurity: Wireshark, Scapy, MIME validation, ClamAV, HMAC signing, secure file handling
AI / Data Science: scikit-learn, OpenCV, Stable-Baselines3 (PPO), deep learning, NLP, image processing
Systems & Tools: Linux, Docker, Git, FastAPI, Celery, Redis, React, Streamlit, SQLite, SSH, Chart.js
Embedded & Hardware: Digital logic design, computer architecture, robotics applications

Achievements, Certifications & Languages

- Cybersecurity & AI Internship Certificate — MAHE–ISAC Centre of Excellence, Manipal Institute of Technology (Sep 2025)
- DeepX AI Competition — Top 30 of 150 teams (top 20%), Nile University (2025)
- Languages: Arabic (native), English (fluent)

Portfolio: dolphiny-1412.github.io